

Report on
Curse of Dimensionality, K-Means Variants, and Matrix Factorization

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by

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1) Explain how the curse of dimensionality affects unsupervised learning algorithms.

→ The "curse of dimensionality" refers to the challenges that arise when analyzing and organizing data in high dimensional spaces. As the number of dimensions (features) increases, the volume of the space grows exponentially, making the data increasingly sparse. This sparsity is problematic for any method that requires statistical significance.

Implications for unsupervised learning:

- (i) Data sparsity: High-dimensional data becomes sparse, making pattern detection and clustering less reliable.
- (ii) Distance metrics breakdown: Points appear almost equidistant, reducing the effectiveness of distance based algorithms like K-means.
- (iii) Overfitting risk: More dimensions can lead to fitting noise instead of true patterns, harming generalization.
- (iv) Increased computation: Higher dimensions require more processing power and resources.

To mitigate these issues, dimensionality reduction techniques like PCA (Principal Component Analysis) are often employed before applying unsupervised learning algorithms.

2) Differentiate between K-Means and Kernel K-Means.

→

Feature	K-Means	Kernel K-Means
Distance Metric	Euclidean distance between points.	Implicit distance in a higher-dimensional feature space.
Cluster Shape	Spherical clusters (linearly separable).	Arbitrary, non-linear clusters.
Centroids	Well-defined centroids in original space.	Centroids exist in feature space, not directly interpretable.
Computational Cost	Lower, as it operates in the original space.	Higher, due to kernel computations and matrix operations.
Use Case	Suitable for linearly separable data.	Suitable for complex, non-linear data structures.

3) Define matrix factorization. Why is it important in dimensionality reduction?

→ Matrix factorization is a technique used to decompose a matrix into a product of two or more smaller matrices. This decomposition helps in uncovering latent structures within the data, facilitating tasks like dimensionality reduction and feature extraction.

Importance in dimensionality reduction:

- (i) Capturing latent factors: Identifies underlying patterns, reducing the number of variables needed.
- (ii) Noise reduction: Filters out noise, retaining the most significant information.
- (iii) Improved performance: Reduces dimensions to speed up computation and enhance model efficiency.

Applications:

- (i) Recommendation systems.
- (ii) Image and ~~text~~ processing.